

GUSTAVO LIMA DE LUCENA

SOFTWARE ENGINEER | BACKEND DEVELOPER

gustavolucenaemprego@gmail.com • +55 (81) 98230-2174 • Pernambuco, Brazil (Remote-ready)
github.com/Gustav-Stack • linkedin.com/in/gustavo-lucena1 • gustav-stack.github.io/Portfolio

PROFESSIONAL SUMMARY

Backend Developer with skills in Node.js ecosystem (NestJS, TypeScript, Prisma) and working knowledge of Python, building secure, scalable APIs with authentication, real-time communication, and containerized deployment. Bachelor's degree in Software Engineering, with academic research on Natural Language Processing fundamentals (embeddings, vector similarity, contextual disambiguation) applied to human cognition. Complements technical expertise with hands-on experience in direct sales and negotiation, bringing a practical understanding of market needs and business strategy to product decisions. Professional working proficiency in English (B2), improving continuously. Looking to tackle complex engineering challenges and apply AI in practical, business-driving ways.

TECHNICAL SKILLS

- **Core:** Node.js (Advanced), NestJS (Advanced), TypeScript (Advanced), Python
- **Data:** PostgreSQL, Prisma ORM, MongoDB, Redis, Qlik, Power BI
- **Real-Time & Infrastructure:** Socket.io, Docker, Docker Compose, CI/CD on VPS, GCP, DigitalOcean
- **Documentation & Version Control:** Swagger/OpenAPI, Git
- **Additional:** Java, Spring Boot

TECHNICAL PROJECTS

DeliveryApp — Delivery Platform

- Architected and built a RESTful API with NestJS and TypeScript, backed by PostgreSQL via Prisma ORM
- Implemented authentication and role-based access control (RBAC) using JWT to secure endpoints and enforce granular permissions
- Improved response times by introducing a Redis caching layer and containerized the full environment with Docker for consistent deployment
- Engineered geographic route calculation using the Haversine formula to optimize delivery distance estimates
- Collaborated within a development team following a structured code review process to ensure code quality

WhyApp — Real-Time Chat Application

- Built a NestJS/TypeScript API with real-time messaging powered by Socket.io
- Integrated chat functionality end-to-end with the frontend, including JWT authentication and Docker-based deployment

ACADEMIC RESEARCH

Memory and Artificial Intelligence: Mnemonic Techniques and Natural Language Processing *Undergraduate research paper — B.S. in Software Engineering, UniCesumar*

- Conducted an interdisciplinary review connecting cognitive neuroscience (hippocampus, spatial memory) with Natural Language Processing fundamentals applied to memorization techniques
- Studied vector word representations (embeddings) and contextual disambiguation mechanisms in language models, referencing Word2Vec and BERT
- Proposed applying generative AI as a support tool for mnemonic techniques (method of loci / memory palace)
- Theoretical/bibliographic scope — established a conceptual NLP foundation without code implementation at this stage

EDUCATION

B.S. in Software Engineering 2022 – 2026 (Completed)

UniCesumar University Center — Distance Learning

LANGUAGES

- **Portuguese** — Native
- **English** — Professional Working Proficiency (B2)

JOB PREFERENCES

- **Work Arrangement:** Remote (open to other arrangements)